

Timestamp 0 Hi everyone and welcome to week four of Human-AI Interaction.

Week four is a new section where we're going to discuss human-centered AI as a design process.

So rather than focusing only on the interface side of things or the human cognition side of things, we're looking at the human part of human-centered AI as the part that actually involves stakeholders in its design. and this section has three parts but today we're going to start by part one where we're going to discuss understanding user needs and defining the right problem.

Okay so to give you a little overview of what we're going to discuss today we're going to look at what is human servant design and basically we're going to say who to involve in the design process and why,

What are the degrees and different types of moments for the user involvement?

We're going to look at some life cycles of your design processes and the four basic activities that most of these life cycles have.

And we're going to see these models for the design process.

Okay, so what is human-centered design? Human-centered design is a process, as I mentioned. It's a way to achieve a goal that is to make a product or a service that is actually human-centered. So it's a process, as it says here, that is focused on

discovering requirements from users or stakeholders.

It shows you how you can design to fulfill those requirements by producing prototypes and then actually evaluating them.

And it's traditionally focused on users and their goals, although now that we talk about AI, a lot of that has changed or is evolving in a way because the technology is also driving a lot of that change rather than only the user requirements and there's actually an interesting part in this process where you need to trade off the conflicting needs between the needs of what the people need and the requirements of what are the of your software or your system has to be able to fulfill

So this is one of the first life cycle models we're going to look at of how to design in a human-centered way. As you can see this has two diamonds, so square diamonds in the center and that's why we call this a double diamond process made by the British Design Council.

So if you see these two diamonds have like two phases each. So first it explains that you can diverge, so your thinking is diverged, is exploratory, and you try to understand the people and discover their needs. Then you converge, the diamond closes back to this point, where you define

the actual problem that you need to solve. And then you go into a second loop where you actually design the thing that you're thinking to do, where you develop or prototype your ideas and then you deliver or other models call that also testing with your partner with your stakeholders so you start from a challenge and it ends up to an outcome similarly this is a one from IDEO which is one of the most famous design organizations in the world but this is less about a formal model of iteration but it describes a trajectory that is kind of typical but when you're trying to solve a design problem and it has three phases the inspiration ideation implementation and then you look into how the your like you have the first design challenge how to get started you diverge in your thinking similarly as we were saying before you have like the opportunity for design where you have this in ideation sessions and you prototype with them as well and then you converge and diverge in that phase and then finally you have the implementation phase where you have like an innovative solution in theory, like an innovative solution to actually to deploy. Sorry, yeah, to deploy.

And this one is from the Stanford D School, which describes about the design thinking process, which is also quite common. And in human computer interaction

module, this is the one we typically teach, where it kind of gives a little bit more detail and describes it in five phases where first you empathize you do things like interview shadowing you're trying to understand the stakeholder or the people that will use your system and it's non-judgmental then you go through a second phase where you define your problem um to understand what are like by using personas or understanding the role objectives or seeing what are the pain points of your clients or stakeholders then you ideate a lot of solutions with hopefully with a team you prototype them you make mock-up storyboards so you try to fail fast and see it as fast as possible like make prototypes as fast as possible in order to test those prototypes and see what works and what doesn't and just a another one this is the one from Google that you might find more familiar with it similarly has kind of three stages where well it has six phases with three diamonds where you understand and define similarly to the design council one you sketch and decide like a middle one where you kind of iterate and you prototype things and then you prototype the final more more uh high level prototyping like high fidelity prototyping and you validate your ideas so this is the one from to do design sprints from google so these are all different

Timestamp 6 minutes models you can see there is a lot of overlap between them but today's focus and today's actual and today's focus but also the next three sessions focus are going to be at looking at this life cycle process and seeing how that maps out when it comes down when we're designing with ai the the videos that you're going to see today though focus on the first two stages of in this model from the d school and the model from the d school it's about defining and empathize In the model from the Design Council is the first diamond diamond, and in the Google Sprint one is also the first diamond.

So what we're going to see is how to discover, empathize, understand our users, our stakeholders, our clients, whatever you want to call them. And then the second part is how, after understanding them, we can define a meaningful problem that with AI or any kind of technology we're working on, this is broader than AI, we can actually try to solve these.

So you might ask at this point, like, why Timestamp 7 minutes should we involve users or clients or stakeholders in the design of our systems, especially when they could be very complicated technological systems?

So there are many reasons, and I'll show that visually also in the next slide, but

let me kind of list
some of them here. So
first, you do like
expectation management.
The people that are
working with you will have
realistic expectations.
They're not going to
be surprised or disappointed
by the end result.
they're going to be actually
trained while they're kind
of included in the process
and there's it's going to
be more about a communication
with them rather than
hype but they also are going
to feel more ownership right
like they're going to have
um like they're going to
be active stakeholders in the
design process rather than
just passive receivers or
clients of them and it can
make this can make a big
difference in the acceptance
source uh or and acceptance
and success of a product.
But let's see that
in a more fun way.
So I like this diagram a
lot, this cartoon a lot,
because it actually represents,
I think, very clearly
what's the issue when you
don't kind of involve,
understand the user needs
in the process. So here
you have a little cartoon,
a little sketch, where
in the beginning you see
like what we dreamed of
at kickoff, and you can
see that there's a fancy
rocket with a lot of
features including kind of
commemorative custom artwork
titanium fins one-way
mirror viewport and that's
like everything you usually

dream of when you start
of a new project and then
you kind of go full in
in terms of technological
Then what you end up
building is usually something
much more simple, right?
Because you cannot kind
of, you don't necessarily
have the time or the
scheduling for everything.
So that's the second panel
here, where she says
like what we settled for
at launch with the antenna
in a way and some fins.
But the real thing is that
what the user actually needed
was just something to get
over this ramp so they
wanted a bicycle to get over
the ramp not a whole rocket
system that actually does
that so that's actually
very it's a it's a lesson
learned right like you don't
want to spend too much time
and effort and budget on
something that is very far
away from what is actually
the your client or your user
or your stakeholders whatever
it is in each case might
actually need to do so if
you over design something
or you don't talk to them
you will never know exactly
what it is that they need
and now when i talk about
involving your your users or
your stakeholders there are
many ways to do that and
there are different levels
of intensity so you can be
like very hands-on and
involve them and make them a
member of the design team
you can do small group or
individual activities like
interviews or focus groups or

something like that you
can also have like online
contributions from thousands
of users like a pre beta test
like a lot of games do
something like that right or you
have an online exchange
feedback exchange systems or
crowdsourcing design ideas
you can do participatory
design where you have different
checkpoints in time where
you involve your users over
in your design process or
you can even do user involvement
after product release
if you have the capability
of doing so by doing a b
testing or understanding
and analyzing your customer
reviews but that's like if
you can afford that to do
that so late or these things
are also to be fair these
things are not exclusive
from one another you can also
do them in parallel okay
so now that we know what we
need to understand what people
need we need to empathize
to understand as much as
possible about their users
their activities the context
in these activities to
understand so what we want to
build within our system and
that synthesis of the
empathizing phase will produce
an initial set of requirements
that form a sound basis
to start designing and
it's crucial to start to to
define a little bit the
difference between the user needs
and the product requirements
so requirements are
statements about intended
product that specifies what it
should do or how it should
perform this has to do with

the product right now it doesn't to do with what the user is looking at it has to do with like how does the Timestamp 12 minutes things that the people have told you translate into product features for instance or product characteristics so we have a you a need which is a user perspectives of the problem to be solved and then it you focus on selection you focus and you make a selection out of that and you look at a requirement which is properties of a system that describes the operations of a system that it needs to afford and it usually embodies at least partially one kind of solution or philosophy to start addressing the needs from the users right and there's many strategic approaches that you can do that so a need can have many different kinds of requirements in the end so i guess what you would be asking right now at this stage is like okay this is all great i understand that we need to figure out to talk to our users we need to understand their needs and then figure out how to make this these requirements but how do we elicit discover understand these needs um and in an ideal world you would sit them down you would sit down with your user and then you would just say can you tell us all your needs but that unfortunately wouldn't work as you can imagine for for many reasons one of

which is that people don't necessarily know what they need or they can't articulate it very clearly you need to kind of infer these problems um in another way like famously um or infamously it's a fake quote as far as i know um they were there's this expression that if i asked people what they wanted they would have said faster horses and this is attributed theoretically to henry ford that was making cars at that point basically trying to imply that um implying that if you ask people what they want rather than saying cars the technological innovation they would just say faster horses because that's what they are aware of that's what they know so we cannot also take users or your stakeholders needs literally right you have to kind of be able to infer behind it what what is being said what is being meant So right if they said faster horses, then as a designer, as a good designer, I would think, okay, what I should focus on is on the faster rather than the horses and then figure out how to build something that makes people move faster. so there are a lot of different techniques and this is part of your readings for this week on how to do this you can observe people in their direct or indirectly their behaviors in their environment for example you can do interviews you can do

questionnaires you can do
ethnography where you embed
yourself in the context
of where the users are or
even digital ethnography
where you might be kind of
doing the similar kind of
thing but online communities
you can develop technology
probes so little
prototypes that you give
the people to be able to
see how they react to the
technology itself and then
learn more about the people
through the prototype that
you have actually developed
you can do field studies
as a participant observer
or you can even do critical
design which is not something
we're going to cover
here but this is something
that you will see in your
in chapter seven of your
book okay so we talked about
the empathizing phase and
how to empathize and how to
go about doing that and
now the next part is the
define the define if you
remember was about how to figure
out from the old information
and quotes and interviews
you might have collected
from your users or your
stakeholders you want to
figure out what is the real
problem that you want to solve
so you can make requirements
for your system so the
defined phase in general
is moving from individual
observations to generalized
insights to help focus on
the right problem and i
want to use this as one of my
favorite examples to explain
a little bit what is looking
at the correct problem

what does looking at the correct problem actually mean so say you were out and you were doing observations in a library or observations in general and then you see this you you get this snapshot of an image so what does this child need um think about it for a second like if you would need to write it down what would be your first intuition about what this child needs Yeah, most probably, like when I do this exercise with students, the most answers I get are like something like a chair, a ladder, a footstool, maybe a shorter bookcase is also something that people say as well.

But these are like a little bit like the problem of a faster horse, right? You need to look a little bit beyond the immediate and basically understand that what this child is looking for is actually easier access to information.

So if you think of the problem as a chair or a ladder, then your solution space is going to be, what does the design of a chair look? What does the design of a ladder look? But if you see the problem as an easier access to information, which is more broad, then the solution space that you can solve for that is much, much wider, right? it includes online platforms that includes the ladder and includes the bookcase it includes a lot of other stuff but then

you have a much wider space of understanding what the requirements of your of your system should be

Hi everyone, I hope you saw the first video of the series and now we're going to go into the second video for this week where we're going to discuss more specific examples of how understanding user needs relate to developing AI.

I think the easiest way to understand why we should involve users and people, generally humans, in the development of AI systems is actually this kind of meme or type of Venn diagram. So it started by an online dialogue where somebody was saying that people that talk about AI are not necessarily the same people that have experience with AI.

And Sasa Kostanzacok, he wrote the Design Justice book, he's a very established researcher, responded with this graph. He said that people working on AI are actually a very, very small red circle in comparison by the people most likely to be harmed by the ways that companies and governments use AI.

Timestamp 1 minute So that actually gives us a strong motivation about why we should try to include participants and include users into the development of these systems, especially for you guys that are the ones that are going to be the future

developers of such systems.

Now, this was very nice to think about in theory, but how do we do that in practice? How do we include participants and humans in the development or non-experts in the development of our expert systems?

I wanted to start with this paper, to present this paper first, as it actually tries to see that. it tries to see how can it involve lay users or anyway people that are not experts basically in ai development to develop human-centric responsible ai systems when should you involve these people and how and the way they try to identify that is by doing an online survey of around a thousand two hundred people in the united kingdom and had kind of series of interesting findings.

First, they realized that offering them this kind of diagram of a of the traditional kind of AI development system, people didn't know necessarily where they could be involved. And they found that they had concerns about fairness and transparency of systems, but that it required more education about AI in general to to be able to involve them in this development system, which basically what I'm trying to summarize here is like the first step if you want to include people in the design of your systems is to help educate them so to build towards a more

general AI literacy but besides the general AI literacy there is also very specific things you can do depending on the moment of the life cycle of a program here we're going to use this process this life cycle of AI to show you some examples specifically of how other researchers and companies have managed to include users along the way.

So let's start from the beginning.

We have early in the process where you're looking at your business case and you're actually trying to see why you should be using AI in the first place. um you want to see you have like two big questions to ask is like what why is ai needed and can it be what can be improved if you're actually using it so you want to see basically if ai adds a unique value and this is kind of like the the documentation from um the recommendations from the google pair system pair playbook which i will share with you in the materials for this week but it talks about two kind of important distinctions that is that i think are very useful to start thinking in the beginning when you're building your business case so you're trying to balance the capabilities of your intended user needs you have a people first approach where you try to identify real problems that our people need and they need help with, similarly to what we discussed

in the previous video.

And the key to this approach is articulating and validating your assumptions with real people and to see how these solutions kind of solve them in context.

But as I also mentioned in the previous video, in AI-specific settings, very often there's a technology-first approach where it starts with the capabilities and limitations of the technology. And through that technology, you try to identify how to solve different problems or address user needs and this is quite a popular approach for AI products that need a lot of under the hood improvements like like making safety classifiers to improve user experience and these are things that the participants your users might not know how to ask for in the first place now both approach can offer great product design that people will enjoy and find useful but it's this balance of trying to like for for human centered AI we have to find this balance between um incorporating what people think and what people understand and observing their behaviors and then also the technology itself so finding this balance amongst the two as a rule of thumb this comes again from the pair with Google guidebook they mentioned that AI is probably better in situations where you need to recommend different content to different users for example like Netflix does to recommend

movies to watch the prediction
of future events like
flight prices for the future
personal personalization
to improve the user
experience like the home
thermostats that might
be personalized based
on user comfort levels
things that have natural
language understanding like
dictation software that
would need kind of ai
to help the gear it um
recognition of entire
entities of of an entity
class of entities sorry an
entire entity of entire
class of entities which is
not kind of possible to
do necessarily manually or
without AI detection of
low occurrence events like
credit card fraud an agent
or a bot experience in a
specific in particular domain
like in a hotel that might
like you might have a bot
in the hotel to help you
do the booking process that
will have a large series
of of users for that and
then also like showing
where dynamic content is more
efficient than a predictable
interface so like a
having ai ai generated
suggestions from a streaming
service for new content that
would be nearly impossible
to find other way but
there are also situations
where you can imagine that
ai is probably not better
so places where you want to
maintain predictability
so that's kind of obvious
that for example a home or
a cancel button is easier
when it always is in the

same place when you want
provide it when you want to
provide static or limited
information then it's
actually kind of an overkill
to use ai when you want
to minimize costly efforts
errors sorry like if the
cost of errors is very
high and that weighs the
benefits of a small increase
in success rate. Exactly,
that's kind of a bit
of an overkill again to
AI when you need complete
transparency in some cases
or when you want to optimize
for high speed and low
cost or high value tests
that are things that are
like, for example, in the
medical domain, you might
not want to be doing that.
okay now let's look at
this in relation to the
previous video where we talked
about the different kind
of techniques to include your
users in the in the process
so in this case coming back
to the reading you have
to do for this week of
interacting designing interactive
systems you might want
to do some observations
with your participants to
see how was this work done
before AI you might want
to do some interviews to
understand people's mental
models about the process or the
models under consideration
so then you will have this
kind of understanding of
the current mental models
and how AI might influence that
you might do questionnaires
you might even do kind
of digital or not digital
like physical ethnography

where you would look at what perceptions the people have of similar products or how this worked on again how was this work done before ai so these are like practical uh techniques or strategies you can use to include your um to understand how participants would how you can include participants in the early stages of development okay and there's actually this tool again coming from pair with google that is actually quite useful because it combines the defined stages as we talked about and the empathize phases quite a lot quite quite nicely so you can try to fill in this form as a way to reflect on these questions you start by making a focus statement of how might we solve the user need and here remember again the problem is more broad than the immediate user need and how can ai solve this in a unique way. And then you kind of go step by step this checklist to help you understand this kind of statement of user needs and the suitability of AI as a checklist. It kind of comes back to the same things that I was talking about before.

So the next step that you might want to include your stakeholders with or your users in are on the data preparation, where you might want to include them in collecting the data but also in cleaning and labeling them.

So for example one way to do that like a key challenge in developing machine

learning systems or AI systems is understanding their performance across a wide range of inputs so what you can do is you can see how different data kind of applies to different groups of users and there are tools like this what-if tool that you can see in the in the link below that is also where to address this challenge researchers from google developed this to analyze ml systems with minimal coding so basically what it does is you test hypothetical situations by testing the data sets on the model for different kind of situations different kind of settings so then you can actually see like you can actually measure like ml fairness metrics and then this is a kind of it's an indirect way of including your users but because you don't necessarily have them do the analysis but it is a way definitely to see how your data might be biased or not towards some kind of situations and you might need to see that you need to collect for example data of equal proportions or need to kind of modify your existing data sets to do this to to make the your system more fair um yeah and another kind of stage of the of the ml cycle would be model building and evaluation where you might allow users to shape the algorithms or develop interactive machine learning approaches that can take late user into account and this is there's quite

a lot of work on this i
will give you some few
examples uh i will give you a
few examples but there's
quite a lot of talk about
explainability and and whatnot
that actually sits on this
stage of the of the ai life
cycle um i wanted to tell
you about this paper which
is already quite um it's
established even though
it's quite old it's quite
early it's quite early in
its time about explainability
and they calls it
explanatory debugging
the debugging
part here is about
allowing users to kind of
fix the errors now the biases
that might exist in your
data set so what it does
is is this they've made a
system to explain users how
the system has made each
of the predictions and then
the it allows the users to
make corrections back to the
learning systems so they
hypothesize the authors
hypothesize that a circle of
explanations and fixes kind
of will allow people to
predict how the system will
behave and thus allow them
to personalize their learning
and kind of fix or align
their mental model closer
to how the the interactive
model works the example
that they're using here is
like predicting where a bunch
of emails should be based
in which folder should be
classified so a kind of
typical example in this case
it would be would it go
into basketball or would it
sorry it would go into hockey

or would it be classified
as baseball so based on
the the emails that you see
here and based on some parameters
that they're explaining
here they they they have
a confidence of where
each email should be placed
and there's an explanation
as you can see on the on
the on the side here of what
was the reasoning for doing
that and there's also um
a way to to i think it's
the f part here the list of
words that are used to make
predictions that you can
actually kind of adjust the
importance that is being
given so you kind of teach the
system and then you explain
and then you go through
this loop um all the time
another way a more how do
you say a more designer way
to approach like the user
participation and model
building and evaluation would
be to co-design or to leverage
particular participatory
or speculative design
methods in this case we will
talk a little bit about that
also in the next session
on prototyping but in this
case you're talking about
making prototypes that might
not necessarily be functional
but allows users to
imagine situations that might
appear in the future and
then you might prioritize
like some interventions and
while making some long-term
strategies or decisions of
requirements for the system
so i i'll give you one
of these papers to have as
additional material i'm not
going to spend too much

time on this but there's one
in the additional material
that you can read more on.
but maybe you would be
at this point saying that
this is more familiar in
cases of classic ML setup,
where the downstream tasks,
so the tasks that are
actually being used for,
are likely to be known.

And you know the users
probably already upfront.

So in participatory ML
approaches might look like the
community that you might
like you might include your
community in these kind of
approaches by giving them
control over the business case
or the problem formulation
the data collector and
the storage the design and
evaluation along this pipeline
however that's not always
this case the especially
in the last few years it's
much more challenging with
a foundation model paradigm
where the downstream use
cases and the stakeholders are
not necessarily connected
from to the model and then

And the point here is
they're not connected both
Timestamp 16 minutes conceptually, so the model is
intending to perform well in
a large set of cases, which
are not bounded, but also
practically, right? The
development primarily happens
within some specific large
-state companies, and they're
not necessarily accountable
to specific kind of cases
or specific kind of domains
or specific communities.

So let's see what that
means in terms of how can
we make participatory

approaches for the paradigm of foundational models. and there's this fascinating paper literally called participation in the age of foundation models that recognizes that there are actually four methods currently this came out in 2024 this might be kind of they did a systematic review and they found four ways of including participants to an extent into the foundation model paradigm so the first one which is perhaps the most technical you might be familiar with is Timestamp 17 minutes having reinforcement learning with human feedback here typically crowd workers or contractors usually you include them in the process without most papers don't say more documentation about who these people are or how they recruited them but this form of feedback is typically through online questionnaires where participants get a compare they have to assign a comparative rating to two model outputs and then the feedback is used of that as a reward function and subsequent reinforcement learning training procedure. So then the participants kind of highlight one or the other options. The second one would be kind of rule sets and policies that they identified. And here this kind of refers to a variety of approaches that try

to synthesize rules
or policies about
the foundation model
behavior based on human
input or deliberation.
so as an example
they mentioned the
anthropics collective
constitutional ai
where they're actually using
um it's called ccai where
um you do there have done
a public polling process to
determine a rule set also
they refer to it also as a
constitution which is a
set of instructions for the
model that reflects specific
values for instance they
might ask to this community
to participants, for example,
a representative sample
of a thousand US adults in
this case, choose a response
that is most respectful.
So the participants vote on
these principles or submit
their own, and those with
high consensus make it
into the rule set that kind
of informs the development
of the foundational model
the responses so another
way for that is red teaming
which was a practice
developed in security computer
security where you would
enlist that you would take
kind of domain experts to
adversariously test systems
to find different kinds
of weaknesses and here
participants are usually
becoming kind of closely embedded
into the team but they're
often chosen for their
expertise their domain
expertise or their technical
expertise so they are a
little bit more inclusive in

terms of participation but they're also very much kind of experts in the systems they're not necessarily um like communities or like this if you remember the original circle of everybody that is impacted of ai it's not necessarily everyone and finally they identified other approaches which is similar to creating rule sets with public input or red teaming but But these are relevant to this particular domain or group. So they might engage with researchers that have engaged with maternal health field or disability and queer communities. So they work very closely with these domains, with this kind of subset of people, communities to kind of do this kind

Timestamp 20 minutes of set of principles to inform the foundation models.

And I'll bring you one example of this participatory design for journalism in the additional reading.

So the interesting thing about these four techniques for participation in foundation models in the age of foundation models is the different kind of scales of inclusion, as they have called it in the paper. so they show that people can either consult so you just have like a short consultation with participants which is basically what the reinforcement learning with human feedback basically does so you do crowd work annotations to improve the output quality you might start including participants

um to an extent with a few more sessions over the workshops for instance For instance, you would have a one-time deliberation session with public stakeholders to produce guidelines for behavior, as in the examples from the general rule Timestamp 21 minutes sets, or in red teaming to do adversarial testing. And then the domain-specific efforts, domain-oriented efforts, seem to be the only ones that come a little bit closer and engage the participants a bit deeper by including them, again, with interviews with marginalized stakeholders. or in the more in the examples that this paper has kind of identified there are more specific cases where there was one specific case where the stakeholders actually created a collaboratively licensed data set received for and they even received kind of payment for the use of the images that was part of that data set when the last kind of level of participation would be owning so what the interesting thing here the reason i'm bringing you this graph is like they have this paper has identified a participatory ceiling where it's much easier to consult or include participants in the beginning and much harder and there's a lack of incentives as they mentioned to share control with these communities so that's might be something that you might want to consider in your practice

in the future of how do
you kind of collaborate collaborate
more or let participants
of communities own what
um what you're actually
developing or co-own obviously
not owned by themselves
okay that was quite a lot
presentation uh theoretical
presentation the both videos
together in the beginning
we described about how the
design research process
functions and we looked at
how we can understand user
needs and why we should do
that and how we can define
the right problem by looking
at example and in the second
part we looked at specific
examples in the ai world
about both traditional ml
processes as well as now
in the foundational model
kind of era how we can include
and how and why we can
include stakeholders in the
design of such systems thank
you and see you next week